

Software Requirements Specification (SRS)

Clinical Trial Management System

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Prepared by: Development Team

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the comprehensive requirements for the Clinical Trial Management System (CTMS). This system will facilitate the end-to-end management of clinical research studies including trial setup, participant recruitment, protocol compliance monitoring, data collection, adverse event reporting, and regulatory submission support while maintaining the highest standards of research integrity, participant safety, and regulatory compliance.

1.2 Scope

The CTMS enables clinical research organizations, pharmaceutical sponsors, principal investigators, and site coordinators to manage comprehensive clinical trial operations from protocol development through study completion and regulatory submission. It supports all stakeholders in participant management, protocol compliance, data integrity, and adverse event management through role-based access control. Key functionalities include:

- Research staff and participant authentication and profile management system
- Role-based access control (Admin, Sponsor, Principal Investigator, Sub-Investigator, Site Coordinator, Data Manager, Participant, Regulatory Affairs Officer)
- Complete CRUD operations for trial protocol and participant management
- Participant recruitment, screening, and enrollment workflow system
- Protocol deviation and amendment management
- Adverse event and serious adverse event reporting system
- Electronic data capture for case report form management
- Investigational product inventory and accountability tracking
- Regulatory document management and submission tracking
- Dashboard interfaces for different research stakeholder roles
- Real-time trial status monitoring and milestone tracking
- Audit trail and data integrity management for GCP compliance

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
API	Application Programming Interface
REST	Representational State Transfer
SRS	Software Requirements Specification
JWT	JSON Web Token
RBAC	Role-Based Access Control

CTMS	Clinical Trial Management System
GCP	Good Clinical Practice
ICH	International Council for Harmonisation
IRB	Institutional Review Board
IEC	Independent Ethics Committee
PI	Principal Investigator
CRF	Case Report Form
eCRF	Electronic Case Report Form
AE	Adverse Event
SAE	Serious Adverse Event
IP	Investigational Product
IND	Investigational New Drug Application
NDA	New Drug Application
CDSCO	Central Drugs Standard Control Organisation
SDV	Source Data Verification
TMF	Trial Master File
SUSAR	Suspected Unexpected Serious Adverse Reaction
JPA	Java Persistence API

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- ICH E6(R2): Good Clinical Practice Guideline
- ICH E3: Structure and Content of Clinical Study Reports
- FDA 21 CFR Part 11: Electronic Records and Electronic Signatures
- CDSCO New Drugs and Clinical Trials Rules 2019
- CTMS Implementation Best Practices Guidelines
- Spring Boot Security Documentation
- React.js Documentation

1.5 Overview

This document provides a comprehensive specification for the Clinical Trial Management System covering all aspects of clinical research operations including trial setup, participant management, data collection, safety reporting, and regulatory compliance. Section 2 describes the overall system. Section 3 details specific functional and non-functional requirements. Section 4 provides supporting appendices including database schemas, API documentation, security configurations, and GCP-compliant business

rules.

2. Overall Description

2.1 Product Perspective

The Clinical Trial Management System is a standalone web application developed using Spring Boot for backend services with Spring Security, React.js for the frontend interface, and secured REST APIs for communication. The system uses JPA for database operations, JWT for authentication, and implements comprehensive role-based access control specifically designed for clinical research operations. The platform integrates with regulatory authority portals, electronic data capture systems, laboratory information systems, and pharmacovigilance databases to provide a complete clinical research management solution compliant with ICH E6(R2) GCP guidelines.

2.2 Product Functions

- Research staff and participant registration with role-appropriate authentication
- Trial protocol setup, versioning, and amendment management
- Participant recruitment, screening, eligibility verification, and enrollment workflow
- Electronic case report form design, completion, and validation
- Adverse event and serious adverse event reporting and tracking
- Investigational product accountability and dispensing management
- Protocol deviation detection, recording, and corrective action tracking
- Regulatory document management and submission milestone tracking
- Site initiation, monitoring visit, and closeout management
- Data query management and source data verification support
- Real-time trial progress dashboard and enrollment metrics
- GCP-compliant audit trail for all data entries and modifications

2.3 User Classes and Characteristics

System Administrator

Full system access including user management, system configuration, audit log oversight, and complete operational control of the CTMS platform.

Sponsor

Trial oversight, protocol approval, safety monitoring committee coordination, interim analysis review, budget management, and regulatory strategy direction.

Principal Investigator

Protocol implementation authority, participant consent oversight, adverse event medical assessment, protocol deviation review, and site team management.

Sub-Investigator

Participant screening and enrollment, clinical assessments, AE documentation, eCRF completion, and IP dispensing under PI oversight.

Site Coordinator

Day-to-day trial operations, participant scheduling, regulatory document management, IP inventory management, and monitoring visit coordination.

Data Manager

eCRF review, data query generation and resolution, database lock management, and data quality oversight with no participant-identifying information access.

Regulatory Affairs Officer

Regulatory submission tracking, document management, authority correspondence, and compliance monitoring without clinical data modification rights.

Participant

Informed consent access, appointment scheduling, symptom diary completion, and study update communications through participant portal.

2.4 Operating Environment

- Server: Spring Boot 3.x with Spring Security and audit logging framework (Java 17+)
- Database: JPA-compatible relational database with clinical trial data tables and audit tables
- Frontend: React.js with role-based authentication components (port 8081)
- Backend: Spring Boot with security and audit trail (port 8080)
- Authentication: JWT-based with 21 CFR Part 11 compliant electronic signature support
- Testing: JUnit for backend, React Testing Library for frontend, validation testing for GCP compliance
- Compliance: ICH E6(R2) GCP, 21 CFR Part 11, CDSCO New Drugs and Clinical Trials Rules 2019

2.5 Design and Implementation Constraints

- 21 CFR Part 11 compliant electronic signatures required for all critical trial data entries
- ICH E6(R2) GCP-compliant audit trail mandatory for all data creation, modification, and deletion
- JWT-based authentication with role-specific access tokens for all protected operations
- Data encryption at rest and in transit mandatory for all participant-identifiable information
- Protocol deviation detection must operate in real-time during eCRF data entry
- Adverse event submission must meet regulatory reporting timelines (SAE within 24 hours)
- System validation documentation required per 21 CFR Part 11 before clinical deployment
- Randomization and blinding management must be implemented with strict access segregation

2.6 User Documentation

- System validation documentation and user acceptance testing protocols
- Role-specific user manuals for investigators, coordinators, and data managers
- GCP training completion tracking and documentation guide

- 21 CFR Part 11 compliance guide for electronic records and signatures
- Regulatory submission tracking and document management guide
- Participant portal usage guide and informed consent access documentation

2.7 Assumptions and Dependencies

- All research staff complete mandatory GCP training before system access
- Integration with institutional IRB/IEC for protocol approval status updates
- Regulatory authority portal access for submission status tracking
- Laboratory information system integration for central lab result integration
- Pharmacovigilance database access for SUSAR reporting and tracking
- Sponsor safety database integration for expedited safety reporting
- Electronic signature certificate infrastructure for 21 CFR Part 11 compliance

3. Specific Requirements

3.1 External Interfaces

3.1.1 User Interfaces

Interface	Description
Landing/Home Page	CTMS platform overview with role-specific login portals
User Registration	Account creation with GCP training verification and role assignment
Login/Authentication	Secure login with electronic signature support and role-based redirection
Sponsor Dashboard	Portfolio view of all sponsored trials, safety signals, and milestone tracking
Investigator Dashboard	Site-level trial status, enrollment metrics, pending tasks, and safety alerts
Coordinator Workbench	Daily task management, participant scheduling, and document tracking
eCRF Module	Electronic case report form entry, validation, and query management
AE/SAE Reporting	Adverse event entry, medical coding, and regulatory reporting interface
IP Accountability	Investigational product receipt, dispensing, and reconciliation tracking
Regulatory Document Hub	TMF management, submission tracking, and authority correspondence

3.1.2 Hardware Interfaces

- Electronic signature pad for participant informed consent capture and documentation
- Barcode scanner for investigational product identification and dispensing verification
- Secure document scanner for source document upload and verification
- Biometric authentication device integration for high-security access to blinded data

3.1.3 Software Interfaces

- Regulatory authority submission portal integration (CDSCO, FDA, EMA) for filing status
- Laboratory information system for central laboratory result retrieval and integration
- Pharmacovigilance database for SUSAR reporting and safety signal monitoring
- Randomization system integration for treatment allocation and blinding management
- Electronic data capture system integration for legacy trial data migration
- Medical coding dictionaries (MedDRA, WHO Drug Dictionary) for AE and medication coding

3.1.4 Communication Interfaces

- HTTPS with TLS 1.3 for all clinical trial data transmissions
- 21 CFR Part 11 compliant electronic signatures embedded in audit trail
- JSON data format with GCP-compliant data integrity headers
- JWT token authentication for all protected endpoints
- Real-time safety alert notifications via WebSocket for SAE reporting

- Encrypted email integration for regulatory correspondence and safety notifications

3.2 Functional Requirements

FR1: Comprehensive User Registration and Authentication System

Description: The system shall provide GCP-compliant user registration with mandatory training verification, role-appropriate credentialing, and electronic signature capability for all research staff accessing the CTMS platform.

Detailed Requirements:

- Registration form shall capture full name, institutional affiliation, professional credentials, role type, GCP training certificate number, and contact information
- GCP training verification shall confirm valid training completion within the past 2 years before account activation
- Principal Investigator registration shall require medical license verification and FDA Form 1572 (or equivalent) acknowledgment
- Electronic signature credentials shall be established during registration per 21 CFR Part 11 requirements
- Role assignment shall be approved by sponsor representative or system administrator before access is granted
- Password security shall enforce complexity requirements with mandatory 60-day expiry for all research staff
- Multi-factor authentication shall be mandatory for all user roles accessing participant-identifiable data
- GCP training expiry tracking shall send renewal reminders 60 days before certificate expiry
- Account deactivation workflow shall be initiated when GCP training expires or staff leaves the study
- Audit trail for all account creation, modification, and deactivation shall be maintained for GCP compliance

FR2: Secure Authentication and Session Management

Description: The system shall implement 21 CFR Part 11 compliant authentication for clinical trial operations, ensuring data integrity, electronic signature validity, and session security throughout research activities.

Detailed Requirements:

- 21 CFR Part 11 compliant electronic signatures shall link each data entry to the authenticated user with timestamp and meaning statement
- JWT-based session management shall provide role-specific access tokens with research-appropriate session durations
- Mandatory MFA for all users accessing unblinded trial data or performing data lock operations
- Session timeout shall occur after 15 minutes of inactivity for participant-identifiable data access

- Electronic signature re-authentication shall be required for critical operations including data lock, protocol deviation sign-off, and SAE submission
- Concurrent session detection shall prevent simultaneous logins for data integrity protection
- Failed login tracking shall implement progressive lockout with administrator notification after 5 failed attempts
- Audit log shall capture every login, logout, session timeout, and failed authentication event with timestamp and IP
- Emergency unblinding access shall be available for patient safety with immediate notification to sponsor and mandatory documentation
- Security certificate management shall track expiry of electronic signature certificates with advance renewal alerts

FR3: Role-Based Access Control for Clinical Trial Operations

Description: The system shall implement GCP-compliant RBAC ensuring appropriate data access, function availability, and audit trail generation for all clinical trial stakeholders based on their role and study relationship.

Detailed Requirements:

- Participant role shall include personal study information access, appointment scheduling, symptom diary completion, and consent document viewing
- Site Coordinator role shall include participant scheduling, regulatory document management, IP accountability, eCRF entry assistance, and monitoring visit coordination
- Sub-Investigator role shall include participant clinical assessment, eCRF completion, AE documentation, IP dispensing, and protocol compliance monitoring
- Data Manager role shall include eCRF review, data query management, database lock operations, and data quality reporting without participant contact information access
- Principal Investigator role shall include full site-level access including consent oversight, AE medical review, protocol deviation assessment, and staff delegation authority
- Regulatory Affairs Officer role shall include regulatory document management, submission tracking, and authority correspondence without clinical data modification
- Sponsor role shall include cross-site trial oversight, safety monitoring, interim analysis access, budget management, and protocol amendment authority
- System Administrator role shall have complete platform access including user management, system configuration, validation documentation, and audit oversight
- Blinding access controls shall prevent unblinded staff from accessing randomization assignments except through emergency unblinding procedure
- Study-specific access shall ensure staff can only access data for trials they are formally enrolled in with documented delegation

FR4: Trial Protocol Setup and Management System

Description: The system shall provide comprehensive trial protocol management enabling structured setup of study parameters, eligibility criteria, visit schedules, assessment requirements, and protocol

version control.

Detailed Requirements:

- Protocol setup shall capture study title, sponsor, phase, therapeutic area, primary and secondary endpoints, and study design parameters
- Eligibility criteria management shall define inclusion and exclusion criteria with automated screening assessment support
- Visit schedule configuration shall define all study visits with windows, required assessments, and mandatory eCRF forms per visit
- Protocol versioning shall maintain complete history of all protocol versions with amendment justifications and effective dates
- Protocol amendment workflow shall route changes through PI review, IRB/IEC submission, and regulatory authority notification as required
- Randomization scheme configuration shall support parallel, crossover, factorial, and adaptive designs with appropriate blinding levels
- IP assignment configuration shall define dispensing schedules, dose levels, and accountability requirements per protocol arm
- Safety stopping rules shall be configured with automated monitoring against predefined thresholds triggering DSMB alerts
- Protocol deviation definition shall capture all parameters that constitute deviations with severity classification
- Multi-site configuration shall manage multiple investigative sites with site-specific contacts, IRB approvals, and regulatory requirements

FR5: Participant Recruitment and Enrollment Management System

Description: The system shall implement comprehensive participant management from recruitment through screening, enrollment, and study completion with full documentation of informed consent and eligibility verification.

Detailed Requirements:

- Participant registration shall create pseudonymized participant records with unique subject IDs preserving blinding requirements
- Informed consent documentation shall capture consent version, date, witness, and method with electronic or scanned signed consent upload
- Screening eligibility assessment shall systematically evaluate inclusion and exclusion criteria with automatic pass/fail determination
- Screen failure documentation shall record reason for screen failure for enrollment analytics and protocol review
- Randomization assignment shall be performed through integrated or linked randomization system with appropriate blinding controls
- Enrollment milestone tracking shall monitor site-level and global enrollment against targets with projections and recruitment trend analysis

- Participant visit scheduling shall manage all protocol-required visits with window calculations and missed visit flagging
- Participant retention management shall track withdrawal reasons, early termination, and lost to follow-up with appropriate documentation
- Re-enrollment management shall handle protocol-defined re-enrollment procedures with appropriate eligibility re-verification
- Participant communication management shall enable study team communication with participants through approved channels per protocol

FR6: Electronic Case Report Form Management System

Description: The system shall provide comprehensive electronic data capture through configurable eCRF forms with real-time validation, discrepancy detection, and GCP-compliant audit trail for all data entries.

Detailed Requirements:

- eCRF form builder shall support configuration of diverse field types including text, numeric, date, dropdown, checkbox, and scale measurement fields
- Visit-specific form assignment shall display only protocol-relevant forms for each scheduled visit type
- Real-time data validation shall check range limits, mandatory field completion, date logic, and cross-field consistency during entry
- Protocol deviation detection shall automatically flag entries that violate protocol-defined parameters with immediate alert generation
- Electronic signature shall be required for eCRF completion with date, time, and user identity captured per 21 CFR Part 11
- Data query workflow shall allow data managers to raise queries on specific fields with site coordinator response and resolution tracking
- Audit trail shall capture original value, modified value, modifier identity, timestamp, and reason for change for every data modification
- Missing data management shall track incomplete fields with configurable follow-up reminders and escalation workflows
- Data import capability shall support laboratory result auto-population from integrated LIS for central laboratory data
- Database lock workflow shall implement data review, query closure verification, and authorized lock with irrevocable audit documentation

FR7: Adverse Event and Safety Reporting System

Description: The system shall implement comprehensive adverse event management from initial report through medical coding, causality assessment, regulatory submission, and follow-up to ensure participant safety and regulatory compliance.

Detailed Requirements:

- AE entry shall capture onset date, description, severity grade, relationship to IP, action taken, outcome, and resolution date
- SAE classification shall identify events meeting serious criteria with automatic escalation and 24-hour reporting timeline enforcement
- SUSAR reporting workflow shall generate expedited safety reports for unexpected serious adverse reactions for regulatory submission
- MedDRA coding integration shall support systematic medical coding of adverse events for regulatory submission requirements
- Causality assessment documentation shall capture investigator and sponsor medical officer assessments with supporting rationale
- Follow-up report management shall track SAEs requiring ongoing follow-up with periodic update capture and resolution documentation
- Safety data reconciliation shall compare site-reported AEs with sponsor safety database for completeness verification
- DSMB safety monitoring support shall provide blinded and unblinded safety data packages for data safety monitoring board review
- Regulatory submission tracking shall monitor SUSAR submission deadlines and confirmation receipt from regulatory authorities
- AE analytics shall provide frequency distributions, severity profiles, system organ class summaries, and temporal trend analysis

FR8: Investigational Product Accountability System

Description: The system shall provide complete investigational product tracking from receipt through dispensing, return, and destruction with full chain of custody documentation for regulatory accountability compliance.

Detailed Requirements:

- IP receipt documentation shall record shipment details, batch numbers, quantity received, condition on receipt, and storage assignment
- IP storage management shall track storage conditions with configurable temperature monitoring alerts for cold chain products
- Dispensing record shall capture participant ID, visit number, dispensed kit number, quantity, dispensing date, and authorizing investigator
- Randomization kit assignment shall link dispensed IP to participant randomization number maintaining appropriate blinding
- IP return and reconciliation shall document returned units, reason for return, condition, and disposal or destruction actions
- IP accountability log shall maintain running balance of received, dispensed, returned, and destroyed quantities per batch
- Drug supply forecasting shall project upcoming IP needs based on enrollment rate and visit schedule to prevent supply gaps

- Expiry management shall flag IP batches approaching expiry with reorder or replacement request workflow
- Temperature excursion documentation shall record any cold chain deviations with quarantine action and sponsor notification
- Final IP reconciliation shall support study closeout with complete accountability documentation for regulatory inspection

FR9: Regulatory Document and TMF Management System

Description: The system shall provide comprehensive regulatory document management supporting the Trial Master File, regulatory submission tracking, IRB/IEC correspondence, and inspection readiness maintenance.

Detailed Requirements:

- TMF structure shall organize documents per ICH E6(R2) TMF reference model with mandatory and optional document sections
- Document upload and version control shall track all regulatory documents with version history, effective dates, and superseded document archival
- IRB/IEC submission tracking shall manage initial submissions, amendments, continuing reviews, and approval status with expiry alerts
- Regulatory authority submission tracking shall monitor IND/NDA filings, clinical trial applications, and authority response management
- Delegation log management shall maintain up-to-date records of PI delegations to study staff with signature and date tracking
- Staff training log shall track GCP training completion, protocol training, and role-specific training requirements for all site staff
- Monitoring visit management shall schedule visits, track findings, capture corrective actions, and manage visit report approval
- Protocol deviation log shall maintain regulatory-required records of all deviations with root cause analysis and CAPA tracking
- Essential document checklist shall verify completeness of required documents at trial initiation, during conduct, and at closeout
- Inspection readiness dashboard shall provide real-time TMF completeness metrics and outstanding document alerts for audit preparation

FR10: Analytics, Reporting, and Compliance Monitoring System

Description: The system shall provide comprehensive analytics supporting trial operational management, safety signal detection, enrollment performance monitoring, and regulatory compliance tracking across all study sites.

Detailed Requirements:

- Enrollment analytics shall display site-level and global enrollment against targets with trend projections and milestone tracking

- Protocol compliance monitoring shall track deviation rates, query resolution times, and eCRF completion timelines by site
- Safety signal monitoring shall provide cumulative AE frequency tables, incidence rates by arm, and temporal trend analysis
- Data quality metrics shall report eCRF completion rates, outstanding query counts, and data error rates by site and form
- IP accountability reporting shall generate regulatory-ready accountability logs by batch, site, and study period
- Timeline management shall display Gantt-style trial milestones with actual versus planned tracking and delay flagging
- Financial analytics shall track site payments, grant disbursements, and budget utilization against study financial plan
- Regulatory submission calendar shall list all upcoming submission deadlines with status and responsible party assignments
- Custom report builder shall allow configuration of cross-study reports for portfolio-level sponsor oversight
- Real-time trial dashboard shall provide at-a-glance status for enrollment, safety, compliance, and operational metrics

FR11: Notification and Communication Management System

Description: The system shall provide comprehensive notifications for trial events, safety alerts, compliance deadlines, and participant communications while maintaining appropriate blinding and data access controls.

Detailed Requirements:

- SAE alert notifications shall immediately contact PI, sponsor medical officer, and regulatory affairs team for all serious adverse events
- SUSAR reporting deadline alerts shall notify responsible parties 48 hours before and at submission deadline
- Enrollment target alerts shall notify sponsor and PI when site enrollment falls behind schedule projections
- IRB/IEC approval expiry alerts shall notify site coordinators and PIs 60 and 30 days before approval expiration
- Protocol deviation notifications shall alert PI and sponsor representative of each deviation with required response timeline
- Participant visit reminder notifications shall remind participants of upcoming scheduled visits via approved communication channels
- Data query notifications shall alert site coordinators of new and overdue data queries requiring resolution
- IP supply alert notifications shall notify coordinators and sponsor when IP stock falls below configured safety levels

- GCP training expiry reminders shall notify research staff 60 days and 30 days before certificate expiration
- Monitoring visit notifications shall inform site teams of upcoming visits with pre-visit document preparation checklists

FR12: System Administration and Configuration Management

Description: The system shall provide comprehensive administrative capabilities supporting system validation, user management, GCP compliance configuration, and operational oversight for the CTMS platform.

Detailed Requirements:

- System validation documentation management shall maintain Installation Qualification, Operational Qualification, and Performance Qualification records
- User account lifecycle management shall handle research staff onboarding, role changes, study assignment, and deactivation with audit trail
- Study configuration management shall allow setup and modification of trial parameters, visit schedules, and eCRF forms per protocol
- Audit log administration shall provide complete searchable audit trail for all system activities for GCP inspection
- System configuration management shall control global settings for session timeouts, password policies, and electronic signature requirements
- Data backup and recovery shall perform encrypted daily backups with tested restoration procedures and off-site storage
- Integration management shall monitor connectivity with laboratory systems, regulatory portals, and randomization platforms
- Validation change control shall manage system updates through formal change control with impact assessment and re-validation as required
- Performance monitoring shall track system response times, database load, and concurrent user capacity
- Disaster recovery planning shall maintain documented recovery procedures with regular drill testing and RTO/RPO compliance

3.3 Performance Requirements

Requirement	Target
Authentication and Electronic Signature	< 2 seconds for 95% of operations
eCRF Load and Save	< 2 seconds for form loading and data save operations
Protocol Deviation Detection	< 1 second from data entry to deviation flag display
SAE Alert Delivery	< 30 seconds from SAE entry to notification delivery
Report Generation	Standard trial reports generated within 15 seconds
Concurrent Users	Support 200+ concurrent research staff across all sites
Database Capacity	Handle 500+ active trials and 1,000,000+ eCRF data points
Audit Trail Retrieval	Audit records accessible within 5 seconds for any data point
Document Upload	Support files up to 50MB with upload completion within 10 seconds
System Availability	99.9% uptime with < 2 hours planned maintenance monthly

3.4 Logical Database Requirements

Core Tables:

Users

id, username, email, passwordHash, role, institutionalAffiliation, gcpCertNumber, gcpExpiryDate, electronicSignatureHash, createdAt, lastLogin, isActive

Studies

id, studyTitle, sponsorId, phase, therapeuticArea, status, protocolVersion, irbApprovalDate, regulatoryApprovalDate, primaryEndpoint, targetEnrollment

Participants

id, studyId, subjectId, siteId, enrollmentDate, armAssignment, status, consentVersion, consentDate, screeningDate

ProtocolVisits

id, studyId, visitName, visitCode, scheduledDay, windowMinus, windowPlus, visitType, isMandatory

eCRFData

id, participantId, visitId, formId, fieldId, fieldValue, enteredBy, enteredAt, modifiedBy, modifiedAt, modificationReason

AdverseEvents

id, participantId, studyId, aeDescription, onsetDate, severity, causality, seriousCriteria, status, reportedBy, reportedAt

IPAccountability

id, studyId, siteId, batchNumber, kitNumber, quantityReceived, quantityDispensed, quantityReturned, quantityDestroyed, currentBalance

ProtocolDeviations

id, participantId, studyId, deviationType, description, discoveryDate, reportedBy, piAssessment, capaDescription, status

RegulatoryDocuments

id, studyId, documentType, documentTitle, version, effectiveDate, submittedTo, submissionDate, approvalStatus, fileReference

AuditLogs

id, userId, studyId, action, entityType, entityId, oldValue, newValue, electronicSignature, ipAddress, timestamp

3.5 Design Constraints

- 21 CFR Part 11 compliant electronic records and signatures mandatory for all critical data entries
- ICH E6(R2) GCP-compliant audit trail required capturing who, what, when, and reason for every data change
- Randomization and blinding management must prevent unauthorized access to treatment assignments
- SAE reporting workflow must enforce 24-hour notification timeline with automated escalation
- All participant data must be stored with pseudonymization separating identifying information from clinical data
- System validation documentation (IQ/OQ/PQ) required before deployment for regulated clinical trials
- Protocol deviation detection must operate in real-time at data entry point without batch processing delays
- Database lock operations must be irrevocable except through documented and authorized unlock procedure

3.6 Software System Attributes

3.6.1 Reliability

- 99.9% uptime with redundant infrastructure to protect active trial data integrity
- Transaction rollback support for all eCRF entry operations preventing partial saves
- Automated database integrity checks for audit trail completeness verification
- Encrypted backups every 4 hours with off-site storage and quarterly restoration testing
- Graceful degradation allowing read-only access to critical data during maintenance

3.6.2 Availability

- 24/7 availability for global multi-site trials spanning multiple time zones
- Geographic redundancy for disaster recovery with documented RTO of 4 hours

- Planned maintenance windows scheduled during lowest global trial activity periods
- Real-time system monitoring with automated incident alerting for operations team
- Scalable architecture supporting portfolio expansion across multiple concurrent trials

3.6.3 Security

- Authentication: JWT with mandatory MFA and 21 CFR Part 11 electronic signatures
- Authorization: Study-specific RBAC with blinding access segregation
- Data Protection: AES-256 encryption for all participant-identifiable data at rest and TLS 1.3 in transit
- Audit Trail: Tamper-evident, GCP-compliant audit log for every data operation
- Compliance: ICH E6(R2), 21 CFR Part 11, and CDSCO Clinical Trials Rules adherence

3.6.4 Maintainability

- Modular architecture separating trial management, eCRF, safety, IP, and regulatory modules
- Minimum 85% code coverage with validation testing protocols for GCP compliance
- Formal change control process with impact assessment and re-validation documentation
- Comprehensive API documentation and integration guides for third-party connections

3.6.5 Portability

- Docker containerization with deployment support for regulated cloud environments
- Database-agnostic JPA implementation for infrastructure flexibility
- Cloud deployment on GxP-compliant AWS, Azure, and GCP environments
- Cross-platform web application compatibility for global research staff access

3.7 Other Requirements

- System Validation: IQ/OQ/PQ validation documentation required before regulated use
- Testing Coverage: 85% minimum code coverage with GCP workflow validation testing
- Security Testing: Penetration testing and 21 CFR Part 11 compliance audit
- Performance Testing: Load testing simulating global multi-site concurrent data entry
- Regulatory Compliance: ICH E6(R2), 21 CFR Part 11, and applicable national regulations
- Accessibility: WCAG 2.1 AA compliance for global research staff accessibility

4. Appendices

Appendix A: Role-Based Permission Matrix

Functionality	Participant	Coordinator	Sub-Inv.	Data Mgr	Reg. Affairs	PI	Sponsor	Admin
View Trial Info	Own	Site	Site	Study	Study	Site	All	All
Participant Registration	–	✓	✓	–	–	✓	–	✓
eCRF Entry	Diary	Assist	✓	Review	–	✓	View	✓
AE Reporting	–	Assist	✓	–	–	✓	View	✓
SAE Submission	–	–	✓	–	–	✓	View	✓
IP Dispensing	–	Assist	✓	–	–	✓	View	✓
Data Query Mgmt	–	Respond	Respond	Create	–	Review	View	✓
Protocol Deviation	–	Document	✓	–	–	Assess	View	✓
Regulatory Docs	–	Upload	–	–	✓	Review	View	✓
Database Lock	–	–	–	✓	–	Approve	Approve	✓
Unblinding Access	–	–	Emergency	Emergency	–	Emergency	Emergency	✓
Manage Users	–	–	–	–	–	–	–	✓
System Configuration	–	–	–	–	–	–	–	✓
Audit Logs	–	–	–	–	–	–	–	✓

Appendix B: Database Schema Details

Studies Table

- id (PRIMARY KEY, AUTO_INCREMENT)
- study_title (VARCHAR(500), NOT NULL)
- sponsor_id (FOREIGN KEY → Users.id)
- phase (ENUM: PHASE_1, PHASE_2, PHASE_3, PHASE_4, PHASE_1_2, OBSERVATIONAL)
- therapeutic_area (VARCHAR(100), NOT NULL)
- study_status (ENUM: SETUP, ACTIVE, SUSPENDED, COMPLETED, TERMINATED)
- protocol_version (VARCHAR(20), NOT NULL)
- target_enrollment (INT, NOT NULL)
- actual_enrollment (INT, DEFAULT 0)
- irb_approval_date (DATE)
- regulatory_approval_date (DATE)
- first_participant_in (DATE)
- last_participant_out (DATE)

Participants Table

- id (PRIMARY KEY, AUTO_INCREMENT)
- study_id (FOREIGN KEY → Studies.id)
- subject_id (VARCHAR(50), UNIQUE, NOT NULL – pseudonymized ID)
- site_id (FOREIGN KEY → Sites.id)
- enrollment_date (DATE)
- arm_assignment (VARCHAR(50) – encrypted, blinded access only)
- status (ENUM: SCREENING, ENROLLED, ACTIVE, COMPLETED, WITHDRAWN, SCREEN_FAILED)
- consent_version (VARCHAR(20), NOT NULL)
- consent_date (DATE, NOT NULL)
- withdrawal_date (DATE, NULLABLE)
- withdrawal_reason (TEXT, NULLABLE)

eCRFData Table

- id (PRIMARY KEY, AUTO_INCREMENT)
- participant_id (FOREIGN KEY → Participants.id)
- visit_id (FOREIGN KEY → ProtocolVisits.id)
- form_id (VARCHAR(50), NOT NULL)
- field_id (VARCHAR(100), NOT NULL)
- field_value (TEXT)

- entered_by (FOREIGN KEY → Users.id)
- entered_at (TIMESTAMP, NOT NULL)
- modified_by (FOREIGN KEY → Users.id, NULLABLE)
- modified_at (TIMESTAMP, NULLABLE)
- modification_reason (TEXT, NULLABLE – required for modifications)
- electronic_signature (VARCHAR(500) – hash of user+timestamp+value)
- query_status (ENUM: NONE, OPEN, RESPONDED, CLOSED)

AdverseEvents Table

- id (PRIMARY KEY, AUTO_INCREMENT)
- participant_id (FOREIGN KEY → Participants.id)
- study_id (FOREIGN KEY → Studies.id)
- ae_description (TEXT, NOT NULL)
- onset_date (DATE, NOT NULL)
- resolution_date (DATE, NULLABLE)
- severity (ENUM: MILD, MODERATE, SEVERE, LIFE_THREATENING, FATAL)
- causality (ENUM: UNRELATED, UNLIKELY, POSSIBLE, PROBABLE, DEFINITE)
- is_serious (BOOLEAN, NOT NULL)
- serious_criteria (VARCHAR(500), NULLABLE)
- outcome (ENUM: RECOVERED, RECOVERING, NOT_RECOVERED, SEQUELAE, FATAL, UNKNOWN)
- reported_by (FOREIGN KEY → Users.id)
- reported_at (TIMESTAMP, NOT NULL)
- susar_submitted (BOOLEAN, DEFAULT FALSE)

Appendix C: Validation Rules Summary

Field	Validation Rules	Error Messages
Subject ID	Required, alphanumeric, unique per study	"Subject ID is system-generated and cannot be manually entered"
Consent Date	Required, must not be future date, must precede enrollment date	"Consent date cannot be in the future" / "Consent must precede enrollment date"
AE Onset Date	Required, must not be future date, must be after start date	"AE onset date cannot precede enrollment date"
Protocol Version	Required, format: vX.X, must match approved protocol version	"Invalid protocol version format"
GCP Certificate Number	Required for staff, alphanumeric, must be valid	"GCP certificate number is required and must be valid"
eCRF Numeric Fields	Must be within protocol-defined reference values	"Values outside protocol-defined range for this field"
Date Fields	Required where specified; valid date format	"Invalid date format required"
Modification Reason	Mandatory for any eCRF data modification	"Reason for modification is required per GCP regulations"
IP Quantity	Required, positive decimal, must not exceed dispensing limit	"Quantity is invalid and within protocol limits"

Electronic Signature	Required for eCRF completion, deviation	Signature is not mandatory for this action"
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Appendix D: Security Implementation Details

JWT and Electronic Signature Configuration

Parameter	Value
Token Expiration – All Staff	8 hours (GCP session requirement)
Token Expiration – Participant Portal	24 hours
Electronic Signature Re-auth Window	30 minutes per critical operation session
Refresh Token	Not implemented – full re-authentication required per GCP
Algorithm	HS256 with minimum 256-bit secret key
Electronic Signature Format	Username + timestamp + action + data hash per 21 CFR Part 11
Audit Trail Hash Chain	SHA-256 hash linking each audit record to previous for tamper detection

21 CFR Part 11 Compliance Measures

- Electronic signatures are legally binding and linked to individual user credentials
- Each signature captures: printed name, date/time of signing, and meaning of signature
- System controls prevent reuse of electronic signatures from one context to another
- Audit trail is computer-generated and cannot be modified or deleted by any user
- Operational system checks enforce sequencing of events for data entry and approval
- Authority checks confirm user authorization for each system function before execution

Blinding and Randomization Security

- Treatment assignment data stored in encrypted separate schema accessible only to unblinded roles
- Emergency unblinding requires two-factor confirmation and immediate sponsor notification
- IP dispensing labels generated without treatment arm identification for double-blind studies
- Unblinding audit trail captures accessor identity, reason, and timestamp with irrevocable record

Exception Handling Classes

```
public class ProtocolDeviationException extends RuntimeException {  
    public ProtocolDeviationException(String message) { super(message); }  
}  
  
public class UnauthorizedUnblindingException extends RuntimeException {  
    public UnauthorizedUnblindingException(String message) { super(message); }  
}  
  
public class SAEReportingDeadlineException extends RuntimeException {  
    public SAEReportingDeadlineException(String message) { super(message); }  
}  
  
public class DatabaseLockViolationException extends RuntimeException {
```

```
public DatabaseLockViolationException(String message) { super(message); }  
}  
  
public class GCPComplianceException extends RuntimeException {  
    public GCPComplianceException(String message) { super(message); }  
}
```

Appendix E: Integration Specifications

Regulatory and Ethics Integration

- CDSCO clinical trial registration portal for mandatory trial registration and status updates
- IRB/IEC online submission portal for protocol approval and continuing review management
- FDA and EMA regulatory portals for IND/CTA submission status tracking
- WHO International Clinical Trials Registry Platform for mandatory trial registration

Clinical Data Integration

- Central laboratory LIS integration for direct result import into participant eCRF data
- Pharmacovigilance database connectivity for expedited SUSAR reporting and tracking
- Randomization and trial supply management system for treatment allocation
- Medical coding dictionaries (MedDRA, WHO Drug Dictionary) for standardized AE coding

Sponsor System Integration

- Sponsor safety database for reconciliation of site-reported AEs with global safety data
- Clinical data management system for final study data export in CDISC SDTM format
- Electronic Trial Master File platform for document synchronization
- Statistical analysis system integration for interim and final analysis data delivery

Appendix F: Business Rules and Constraints

Participant Management Rules

- Informed consent must be documented before any study procedure is performed
- Subject IDs must be system-generated and pseudonymized to maintain participant privacy
- Screen failure documentation is mandatory and must include specific eligibility criterion violated
- Withdrawal from study must be voluntary; withdrawal reason documentation is required
- Protocol-defined re-screening intervals must be enforced before allowing re-enrollment

Data Integrity Rules

- All eCRF data modifications must include reason for change per ICH E6(R2) GCP
- Database lock is irrevocable without documented and sponsor-approved unlock procedure
- Audit trail entries cannot be deleted or modified by any user including administrators

- Electronic signatures cannot be reused across different forms or time periods
- Source data verification must be completed before database lock at study closeout

Safety Reporting Rules

- All SAEs must be reported to sponsor within 24 hours of site awareness
- SUSARs must be submitted to regulatory authorities within 7 or 15 days per reporting category
- PI must provide medical assessment for all AEs reported at their site
- Follow-up reports required for all unresolved SAEs until resolution or study end
- AE causality assessment must be performed by medically qualified investigator

Data Retention Rules

- All clinical trial data must be retained for minimum 15 years after study completion
- Audit trail records must be maintained for the full data retention period
- IP accountability records retained for minimum 5 years post-marketing approval or discontinuation
- Regulatory documents retained per applicable regulatory authority requirements (minimum 15 years)
- Participant consent forms retained for minimum 15 years after study completion

Appendix G: Future Enhancement Roadmap

Phase 2 Features

- EDC system native mobile application for remote data entry by site staff
- Automated CDISC SDTM and ADaM dataset generation for regulatory submission
- Risk-based monitoring dashboard with centralized statistical monitoring algorithms
- Participant mobile application for ePRO data collection and visit reminders
- Integration with wearable devices for continuous remote patient data collection

Phase 3 Features

- AI-powered protocol feasibility assessment and site selection analytics
- Machine learning recruitment prediction and patient identification from EHR data
- Automated safety signal detection using Bayesian statistical methods
- Natural language processing for medical coding automation
- Adaptive trial design support with real-time interim analysis integration

Phase 4 Features

- Decentralized clinical trial platform with remote consent and home visit management
- Blockchain-based immutable audit trail for GCP compliance assurance
- AI-powered clinical study report generation from trial data
- Digital twin modeling for virtual control arms in confirmatory trials

- Real-world evidence integration for hybrid clinical trial designs

Appendix H: API Summary

Trial and Participant Management

Method	Endpoint	Description
POST	/api/studies	Create new clinical trial setup
GET	/api/studies	Get all studies with filters
GET	/api/studies/{id}	Get study details and protocol
PUT	/api/studies/{id}	Update study parameters
POST	/api/participants	Register new study participant
GET	/api/participants/{id}	Get participant study record
PUT	/api/participants/{id}/status	Update participant status
POST	/api/participants/{id}/consent	Record informed consent

eCRF and Safety Management

Method	Endpoint	Description
GET	/api/ecrf/{participantId}/{visitId}	Get eCRF forms for visit
POST	/api/ecrf/data	Save eCRF data entry with e-signature
PUT	/api/ecrf/data/{id}	Modify eCRF data with reason
POST	/api/adverse-events	Report new adverse event
GET	/api/adverse-events/study/{studyId}	Get all AEs for study
PUT	/api/adverse-events/{id}/sae	Escalate to SAE with reporting
POST	/api/queries	Create data query on eCRF field
PUT	/api/queries/{id}/respond	Respond to data query

IP and Regulatory Management

Method	Endpoint	Description
POST	/api/ip/receipt	Record IP shipment receipt
POST	/api/ip/dispense	Record IP dispensing to participant
GET	/api/ip/accountability/{studyId}	Get IP accountability report
POST	/api/documents	Upload regulatory document to TMF
GET	/api/documents/{studyId}	Get TMF documents for study
GET	/api/admin/audit-logs	Access GCP-compliant audit trail
GET	/api/admin/analytics	Trial analytics and enrollment data

GET	/api/admin/system-health	System status and performance
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Appendix I: Component and Navigation Structure

React Components

App.jsx

- Main routing component with React Router v6 and study-specific route protection
- Routes: '/' (Home), '/login' (Login), '/dashboard' (Dashboard), '/studies' (Studies), '/participants' (Participants), '/ecrf/:participantId/:visitId' (eCRFEntry), '/adverse-events' (AEReporting), '/ip-accountability' (IPAccountability), '/regulatory' (RegulatoryHub), '/reports' (Reports), '/admin' (AdminPanel)
- BlindedRoute wrapper prevents unblinded data access for blinded study staff
- Study-specific context provider supplies active study parameters to all child components

NavBar.jsx

- Application title: 'Clinical Trial Management System'
- Study selector dropdown for users with access to multiple trials
- GCP training expiry warning indicator for current user
- Real-time safety alert badge for unacknowledged SAE notifications

Home.jsx

- Welcome message: 'Welcome to the Clinical Trial Management System'
- Description: 'GCP-compliant trial management for sponsors, investigators, and site teams'
- Navigation buttons: 'Staff Login', 'Participant Portal Login'
- Platform statistics: active trials, enrolled participants, pending safety reports

eCRFEntry.jsx

- Form title: 'Electronic Case Report Form – [Visit Name]'
- Field rendering based on form configuration with real-time validation
- Protocol deviation alert panel when entered data triggers deviation rules
- Modification reason capture modal for any change to previously saved data
- Electronic signature dialog before final form submission
- Audit trail viewer showing complete history of field-level changes

AEReporting.jsx

- Page title: 'Adverse Event Reporting'
- AE entry form with MedDRA coding integration
- SAE classification checklist with automatic escalation trigger
- Timeline display of AE course with follow-up entry capability

- SUSAR reporting status with deadline countdown for serious events

Footer.jsx

- Copyright text: '© 2024 Clinical Trial Management System. All rights reserved.'
- GCP compliance notice and regulatory compliance statement
- Included in all authenticated research staff components

Document Version Control

Version	Changes	Author
1.0	Initial SRS document creation for Clinical Trial Management System	Development Team